

**SCHOOL OF ENGINEERING  
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY**

**B.TECH DEGREE II SEMESTER FIRST INTERNAL EXAMINATION, APRIL 2024**

Sem: S2 EC/ IT (A&B Batches)

Course Title: **23-200-0203B : DIGITAL ELECTRONICS (2023 Scheme)**

Faculty: Dr Deepa Sankar, Dr Binu Paul

Time: 2Hrs

Max. Marks: 50

**Course Outcomes** On successful completion of teaching-learning and valuation activities, a student would be able to

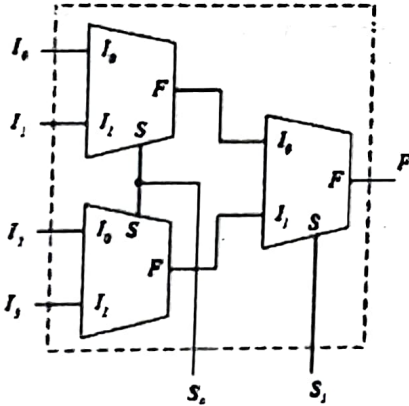
CO1 Understand Boolean postulates and Basic building blocks of Digital systems

CO2 Design Optimal digital circuits using basic building blocks

CO3 Analyse basic digital circuits

CO4 Understand HDL models of simple circuits

L1: Remember, L2: Understand, L3: Apply L4: Analyse L5: Evaluate L6: Create

| Qn<br>no                                                                            | <b>PART A (Answer ALL questions: marks 5 x 1 = 5)</b>                                                                                                                                                                                       | CO, PO, BL |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1                                                                                   | Which of the following number systems uses the symbols 0-7 ?<br>(a) Binary (b) Octal (c) Decimal (d) Hexa decimal                                                                                                                           | 1,(123),1  |
| 2                                                                                   | In Boolean algebra, what is the result of applying De-Morgan's Theorem to the expression $\overline{A + B}$ is<br>(a) $\overline{A} + \overline{B}$ (b) $\overline{A} \cdot \overline{B}$ (c) $\overline{A} + \overline{B}$ (d) $A \cdot B$ | 1,(123),1  |
| 3                                                                                   | Identify the circuit, given that modules shown are 2:1 MUX.<br>(a) 2:1 MUX (b) 4:1 MUX (c) 3:1 MUX (d) None of these                                                                                                                        | 3,(1234),4 |
|  |                                                                                                                                                                                                                                             |            |
| 4                                                                                   | Which gate can be used to implement ANY Boolean function?<br>(a) AND (b) OR (c) NOR (d) XNOR                                                                                                                                                | 1,(123),1  |
| 5                                                                                   | Which gate can be used to implement 2 bit equality detector?<br>(a) AND (b) OR (c) NOR (d) XOR                                                                                                                                              | 1,(123),1  |
| Qn<br>no                                                                            | <b>PART B (Answer ANY 5 questions: marks 5x3 = 15)</b>                                                                                                                                                                                      | CO, PO, BL |
| 6                                                                                   | Express X = 5 in 4 bits and then perform the following arithmetic                                                                                                                                                                           | 1,(123)2   |

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|              |                                                                                                                                                                                                             |                  |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
|              | i) X-6 using 1's complement method<br>ii) 7-X using 2's complement method<br><b>Explain your answers</b>                                                                                                    |                  |
| 7            | Design a circuit to add 4 bits (A,B,C,D) using basic gates (AND, OR, NOT). Provide the truth table for the adder circuit. Clearly illustrate the design steps.                                              | 2,(123)3         |
| 8            | Design a circuit to convert 3 bit binary to its equivalent gray. Show the truth table and design steps. What is the speciality of gray code.                                                                | 2,(1234),3       |
| 9            | Simplify the Boolean expression $F(A, B, C) = \Sigma(m_0, 1, 2, 4, 5)$ . Show the steps involved in the minimization process using Karnaugh maps.                                                           | 2,(123)4         |
| 10           | Discuss how a half subtractor works. Design the circuit and draw it.                                                                                                                                        | 1,(123)2         |
| 11           | A door is having a binary string as its pass key. Design a circuit to open a door if the binary pass word entered is BCD code of 89 d. State your assumptions clearly.                                      | 2,(123)3         |
| <b>Qn no</b> | <b>PART C (Answer All: marks 3 x 10=30)</b>                                                                                                                                                                 | <b>CO ,PO,BL</b> |
| 12           | Discuss how a full adder / full subtractor with mode control can be designed, with relevant design steps and circuit. The circuit should behave as adder for Mode=0 and subtractor for mode =1<br><b>OR</b> | 2,(123)3         |
| 13           | Explain the working of BCD adder ( single digit) with proper design steps.                                                                                                                                  | 2,(123)3         |
| 14           | Reduce and implement following function F, in NAND - NAND way for SOP implementation $F = \Sigma m(0,2,3,8,9,10,11,13,15)$<br><b>OR</b>                                                                     | 2,(123)4         |
| 15           | Design a digital circuit to glow 3 LEDS, to show whether a given 4 bit number is a valid XS3, valid BCD , valid radix 12 number.                                                                            | 2,(123)4         |
| 16           | Implement a 4:1 MUX using basic gates. What modification is needed to give an active high and active low enable for the design.<br><b>OR</b>                                                                | 2,(123)3         |
| 17           | Discuss the internal circuit of a simple 3:8 decoder. How can it be used to implement $F = \Sigma m(2,4,5,7)$ Explain your circuit                                                                          | 2,(123)3         |

|     | 1   | 2   | 3   | 4   |
|-----|-----|-----|-----|-----|
| CO% | 12% | 87% | 1%  | 0%  |
| BL% | 5%  | 7%  | 59% | 29% |